
**Hierarchical structuring of weapon allometry in the
European earwig**

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Abstract

Static allometries are often treated as stable descriptors of morphological architecture, yet scaling relationships themselves may evolve and respond plastically to environmental conditions. We quantified sex- and morph-specific weapon allometries (focusing on the forceps, which are hardened rear appendages used for defence, hunting, and mating) in the European earwig (*Forficula auricularia*) under experimental manipulation of developmental diet and density using hierarchical models that allow scaling exponents to vary across biological levels. Forceps length scaled positively with body size in all groups, but males exhibited substantially steeper allometric slopes than females. There was some evidence that nutritional stress was associated with reduced male scaling exponents, although support for these effects was variable and context-dependent, while density had little consistent influence on slopes. Female scaling relationships remained comparatively stable across environments. Among males, macrolabid individuals tended to show greater environmental sensitivity in scaling relationships than brachylabid males, although statistical support for these differences was limited. There was little evidence that environmental or morph-specific differences were driven by intercept shifts independent of body size. Among-family variation in slopes and intercepts was modest, and their covariance was weak and uncertain. Overall, these results suggest that weapon allometry is structured primarily through variation in scaling exponents and provide qualified support for the idea that slopes may represent condition-sensitive components of exaggerated trait architecture.

Introduction

Static allometry describes how trait size covaries with body size among individuals at the same developmental stage and provides a central framework for understanding the evolution of exaggerated morphology. Sexually selected weapons often exhibit positive allometry, such that larger individuals tend to express disproportionately larger traits (Kodric-Brown et al. 2006; Bonduriansky 2007a; Voje 2016). However, hyperallometry is not universal. When traits serve dual sexual and viability functions, or when functional and mechanical constraints limit disproportionate investment, isometry or hypoallometry may evolve instead (Bonduriansky and Day 2003; Eberhard et al. 2018; Kelly 2026). Consequently, scaling relationships reflect how developmental processes, functional performance, and selection are associated with patterns of trait variation.

Static allometry is typically described by a linear relationship between log-transformed trait size and body size. In this framework, the slope (scaling exponent) describes the strength of covariation between trait size and body size (i.e., size-dependent exaggeration), whereas the intercept (elevation) represents expected trait size at a reference value of body size (e.g., the mean when body size is centred). Static allometry can therefore diverge through changes in either parameter. Differences in scaling exponent reflect variation in the strength of covariation between trait size and body size, whereas intercept divergence reflects differences in expected trait size at that reference body size. These components need not evolve together (Lande 1980; Cheverud 1982; Pélabon et al. 2014). Sexual dimorphism, for example, may arise because males and females differ in scaling sensitivity, in expected trait size, or in both. Similarly, alternative male morphs may diverge through elevation shifts while retaining similar scaling exponents, or

through differences in scaling relationships (Tomkins et al. 2005). Distinguishing between slope-based and intercept-based divergence is therefore essential for understanding how exaggerated traits evolve (Bonduriansky 2007a; Shingleton and Frankino 2013; Pélabon et al. 2014; Voje 2016).

Species expressing alternative reproductive phenotypes provide particularly valuable models for examining the structure of allometry. In these species, males adopt discrete morphs that differ in weapon size, behaviour, and mating tactics (Gross 1996; Shuster and Wade 2003; Oliveira et al. 2008). Because morphs share much of their genome and developmental environment, differences in scaling parameters can reveal how exaggeration varies within a single species. However, empirical analyses of morph-specific static allometry remain scarce, and even fewer studies have investigated how environmental factors influence both slopes and intercepts in polymorphic systems (Kochensparger et al. 2024; Kelly 2026).

Environmental conditions can influence trait and body size variation and consequently reshape static allometries (Emlen 1997; Frankino et al. 2005; Shingleton et al. 2007). Nutritional stress may reduce expected trait size at a given body size, leading to lower intercepts, whereas social factors such as rearing density could alter patterns of trait expression relative to body size.

Environmental effects on scaling exponents would indicate plasticity in the relationship between trait size and body size, whereas consistent slopes across environments would imply canalisation of scaling relationships (Frankino et al. 2005; Shingleton et al. 2007; Nijhout and German 2012). Determining whether environmental variation modifies intercepts, slopes, or both is therefore crucial for evaluating the flexibility of trait exaggeration. Complementary structural modelling of the same study system demonstrates that environmental effects on forceps size are transmitted

both indirectly via body size and directly through trait-specific processes, particularly in males (Ozkan and Kelly submitted).

Beyond sex, morph, and environment, variation among maternal families of related individuals offers insights into the structure of scaling relationships. Intercept and slope may covary across lineages if shared genetic or developmental factors influence both expected trait size and the strength of covariation with body size (Lande 1980; Hansen and Houle 2008; Klingenberg 2008; Pélabon et al. 2014). A positive intercept–slope covariance indicates that families producing relatively larger weapons at average body size also exhibit steeper scaling relationships, reflecting coordinated variation in trait expression (Bonduriansky and Day 2003; Hansen and Houle 2008; Pélabon et al. 2014). Conversely, independence between parameters suggests modularity or relative decoupling of these components.

The European earwig (*Forficula auricularia*) provides a powerful system for integrating these perspectives on static allometry. Males express discrete forceps morphs that vary considerably in size and curvature, commonly referred to as macrolabial and brachylabial forms (Simmons and Tomkins 1996; Tomkins et al. 2005) (Figure 1). Forceps function both as sexual weapons during male–male interactions and as defensive structures, making them traits with dual roles (Kamimura 2014; Meunier 2024). Previous work has documented morph-specific differences in forceps exaggeration and scaling, suggesting that divergence occurs not only in absolute size but also potentially in scaling relationships. Nonetheless, most studies have primarily compared static slopes, with less attention to intercept divergence, environmental sensitivity of scaling parameters, and variation among families.

Here, we quantify sex-specific and morph-specific static allometries of forceps length in *F. auricularia* under experimentally manipulated diet and density. Rather than treating the scaling exponent as fixed, we test whether it varies across biological levels and environmental conditions. Specifically, we investigate whether sexual dimorphism in weapon exaggeration results from divergence in the scaling exponent, intercept, or both; whether macrolabial and brachylabial males differ mainly in expected trait size or in the strength of covariation between forceps size and body size; and whether larval environmental conditions influence intercepts, slopes, or their covariance. Finally, by estimating variation among families in intercepts and scaling exponents, we assess whether these components of scaling relationships covary across shared backgrounds. Through integrating sex, morph, environment, and family components within a unified hierarchical framework, we evaluate whether static allometry is best characterised as a stable descriptor or as an emergent property of multiple sources of variation (see also Simmons and Tomkins 1996; Tomkins et al. 2005; Kelly 2026).

Methods

Study species and experimental design

Female ($n = 115$) *Forficula auricularia* were hand-collected at night from vegetation in Parc Outremont (Montréal, QC, Canada) during September and October 2024. Individuals were transported to the laboratory at the Université du Québec à Montréal and placed individually in 90-mm Petri dishes filled with moist sand. Each dish was provisioned with a cotton-plugged water tube, and TetraMin Tropical Flakes (United Pet Group, Inc., Germany), and each female was assigned a unique ID code. Earwigs were maintained in the dark at 10 °C in an incubator

(Percival Scientific, Perry, IA, USA) to mimic overwintering conditions. Dishes were checked daily for egg laying, and food and water were replenished as needed. Food and water were removed at the time of egg laying (approximately early December), after which dishes were checked weekly. Starting January 2025, dishes were checked daily for egg hatching. The date of egg hatching was recorded; hatching occurred between 13 January and 29 January 2025, and all 115 females produced hatchlings.

For two weeks following hatching, families were provisioned with a water tube and food (TetraMin Tropical Flakes: crude protein: 46%, crude fat: 11%, crude fibre: 3%). This interval allowed nymphs to complete a moult and reach a size sufficient to withstand manual transfer to experimental dishes. To manipulate the developmental environment, we varied two ecological factors: nutritional condition (diet quality) and social density. Diet treatment consisted of two levels: a “good” diet (100% TetraMin Tropical Flakes), providing high nutritional quality, and a “poor” diet (75% TetraMin Tropical Flakes, 25% non-digestible α -cellulose), providing reduced nutritional content. Density treatment consisted of three levels: low density (one individual per Petri dish), medium density (4 individuals), and high density (8 individuals). These treatments allowed us to examine both nutritional and social environmental effects on growth and morphology. We used a split-brood experimental design in which offspring were randomly assigned to diet and density treatments within maternal families to minimise confounding between maternal background and environmental treatment. Experimental dishes were maintained at 20°C to mimic post-winter developmental conditions and contained a water tube, treatment-specific food, and a refuge (four small pieces of paper straw glued together). A total of 1,265 Petri dishes were prepared. Females (mothers) were euthanized by freezing at -20 °C after

offspring had been assigned to treatments. Dishes were checked several times per week for mortality and sexual maturation, and food and water were replenished as needed.

Morphological measurements

Once offspring eclosed to adulthood, they were removed from their Petri dish, the date was recorded, and they were assigned a unique ID code. Pronotum length (a proxy for overall body size) and forceps length were photographed under a Leica S6D stereo microscope (Leica Microsystems Inc., Concord, ON, Canada) at 0.68× magnification using Enersight software, which automatically adds a scale bar to each image. Measurements were taken to the nearest 0.001 mm (following Lieshout and Elgar 2009) using Fiji image analysis software (Schindelin et al., 2012). All measurements were taken blind to treatment where possible. To assess repeatability, a subset of 30 individuals was measured twice. Intraclass correlation coefficients indicated high repeatability (pronotum width ICC = 0.94; forceps length ICC = 0.97). Log-transformed pronotum width was mean-centred before analysis to facilitate interpretation of intercepts and reduce collinearity among predictors.

Data preparation

All analyses were conducted in the R Statistical Environment [version 4.4.2; R Core Team (2025)]. Data were processed using functions from the tidyverse (e.g., dplyr), including recoding, filtering, and grouping operations. We retained individuals from experimental dishes that met predefined density criteria to preserve the integrity of density treatments (low density: one survivor; medium density: ≥ 3 survivors; high density: ≥ 6 survivors). Individuals lacking sex information were excluded. Forceps length and pronotum width were log-transformed to linearize allometric relationships and reduce heteroscedasticity. Log-transformed pronotum length was not

centred prior to analysis. As a result, model intercepts represent expected forceps length when log-transformed pronotum width equals zero and are therefore not directly interpretable as trait values at a biologically typical body size. Good diet and low density were treated as reference levels throughout. Female sex and brachylabid males served as reference categories where appropriate. Maternal identity was included as a random effect (grouping factor) in all models, with both random intercepts and random slopes for body size.

Male morph classification

Male morphs (brachylabid and macrolabid) were classified using finite mixture modelling implemented in the R package *mixsmsn* [version 1.1.12; Prates et al. (2013)]. We modelled the distribution of raw male forceps length as a two-component skew-normal mixture using the function `smsn.mix()`. Forceps length was analysed without prior size correction so that morph classification reflects emergent weapon phenotype rather than residual variation after accounting for body size. Models were estimated using the expectation–maximization algorithm with automatic initialization (maximum 1000 iterations). Support for bimodality was evaluated using AIC and BIC. Posterior probabilities of component membership were calculated for each male, and individuals were assigned to morphs based on maximum posterior probability. All individuals were retained for subsequent analyses, avoiding arbitrary thresholds and preserving variation near the morph boundary.

Sex-specific allometry

We analysed sex-specific and morph-specific allometries using two complementary models to address distinct biological questions (see Supplementary Material for statistical model equations). The sex-specific model evaluates overall differences between females and males in scaling

relationships and their environmental sensitivity. Because males comprise two discrete morphs that differ in trait expression, this analysis effectively captures average differences between females and the male population as a whole, while allowing slopes to vary across predictors through interaction terms. The morph-specific model, restricted to males, explicitly tests for differences between brachylabic and macrolabic individuals in scaling parameters and their environmental sensitivity. This two-model approach therefore separates broad sex-level comparisons from within-male heterogeneity, avoiding the need to impose a single structure that simultaneously accommodates sex differences and morph variation within males. Results from the sex-specific model should thus be interpreted as averages across male morphs, whereas the morph-specific model resolves variation within males.

To quantify sex differences in allometric scaling and their environmental sensitivity, we fitted a hierarchical Gaussian model using the function `brm()` from the *brms* package (via the *cmdstanr* backend) [version 2.23.0; Bürkner (2017)]. Log-transformed forceps length was specified as the response variable. Fixed effects included log pronotum length (body size), sex, diet, density, and all corresponding interactions. Interactions between body size and other predictors allow the allometric slope (scaling exponent) to vary across sexes and environmental conditions. Main effects and lower-order interactions describe differences in intercepts; however, because body size was not centred, intercepts correspond to trait values at log body size equal to zero and are not directly biologically meaningful. Three-way and four-way interactions involving body size were included to capture potential sex-specific environmental modulation of scaling relationships. Maternal identity ($n = 110$ families) was included as a random intercept and random slope for body size, allowing families to vary in both intercepts and slopes.

Morph-specific allometry

To examine scaling architecture within males, we fitted a second hierarchical model restricted to males using `brm()`. Log-transformed forceps length was modelled as a function of body size, male morph (macrolabial versus brachylabial), diet, density, and all corresponding interactions. Body size $\times$ morph interactions quantify differences in scaling exponents between morphs, while lower-order terms describe differences in intercepts. Higher-order interactions with environmental variables allow both slopes and intercepts to vary across conditions in a morph-specific manner. As morph classification was based on absolute forceps expression, this analysis evaluates differences in scaling architecture among emergent weapon phenotypes without imposing size correction prior to morph assignment. Maternal identity ($n = 104$ families) was again included as a random intercept and slope for body size.

Priors and model fitting

Models were fitted in a Bayesian framework using `brms`. Weakly informative priors were specified for fixed effects and variance components to regularize estimates while remaining non-restrictive on the log scale. Models were fit using Hamiltonian Monte Carlo via `cmdstanr`. Convergence and sampling adequacy were assessed using standard diagnostics ($\hat{R}$, effective sample sizes, and trace plots).

Estimation of scaling exponents and plasticity

Condition-specific allometric slopes were derived from posterior draws using the function `emtrends()` from the *emmeans* package [version 2.0.1; Lenth et al. (2019)]. Environmental plasticity in scaling was quantified as deviations in slope from the reference condition (good diet, low density). We report posterior medians and 95% credible intervals for all slopes, intercepts,

and their contrasts. To aid interpretation, we also calculated posterior probabilities that slope differences exceeded zero and that their absolute magnitude exceeded 0.1 on the log–log scale. All inference is based on posterior distributions rather than null-hypothesis significance testing.

Results

Forceps length scaled positively with pronotum length across all individuals, confirming overall positive allometry (Table S1). Descriptively, females exhibited a continuous distribution of weapon sizes across body sizes ($\bar{x} \pm \text{SE}$ pronotum length = 1.77 ± 0.005 mm; $\bar{x} \pm \text{SE}$ forceps length = 3.59 ± 0.008 mm, $n = 935$), whereas males formed two discrete clusters corresponding to brachylabic ($\bar{x} \pm \text{SE}$ pronotum length = 1.66 ± 0.006 mm; $\bar{x} \pm \text{SE}$ forceps length = 4.064 ± 0.014 mm, $n = 491$) and macrolabic morphs ($\bar{x} \pm \text{SE}$ pronotum length = 1.80 ± 0.006 mm; $\bar{x} \pm \text{SE}$ forceps length = 5.93 ± 0.033 ; $n = 488$; Figure 2).

Sex differences in scaling relationships

Allometric slopes differed strongly between sexes. Because males comprise two morphs with differing scaling relationships, sex-specific results should be interpreted as averages across male phenotypes. Females showed a moderate positive relationship between body size and forceps length (logP: $\beta = 0.35$, 95% CrI: 0.07 – 0.65; Figure 3a), whereas males exhibited substantially steeper scaling (logP $\times$ sex: $\beta = 0.61$, 95% CrI: 0.25 – 0.98; Table S1). This divergence is evident in Figure 3, where male scaling relationships are consistently steeper than those of females across conditions. There was little evidence for consistent differences in intercept terms associated with sex (sex: $\beta = -0.06$, 95% CrI: -0.28 – 0.15). However, because body size was not centred, intercepts correspond to expected trait values at log body size equal to zero and are not directly

biologically interpretable. Accordingly, differences between the sexes are best understood in terms of divergence in scaling slopes rather than shifts in intercepts.

Including a quadratic term for body size provided no meaningful improvement in model fit (see Supplementary Material for statistical model equation). Although the quadratic model showed a slightly higher expected log predictive density, the difference was small relative to its standard error ($\Delta\text{elpd} = 3.7 \pm 3.8$). In addition, credible intervals for the quadratic term overlapped zero, indicating no clear evidence of curvature in the size–trait relationship. We therefore retained the linear model for all subsequent analyses.

Environmental effects on scaling

Environmental effects on allometry were generally weak. In females, scaling relationships were largely parallel across diet and density treatments (Figure 3a), and interactions between body size and environmental variables had 95% credible intervals that overlapped zero (Table S1), indicating little evidence of environmentally induced changes in scaling exponents. In males, there was some indication that the nutritional environment may influence scaling relationships. The interaction between body size, sex, and diet was negative ($\log P \times \text{sex} \times \text{diet}: \beta = -0.68$, 95% CrI: $-1.25 - -0.11$), consistent with a reduction in the steep male allometric slope under poor dietary conditions. This pattern is visible in Figure 3c, where slopes under a poor diet appear shallower than under a good diet, particularly at larger body sizes. However, credible intervals for several condition-specific slopes and derived contrasts overlapped zero (Table S1), indicating that support for diet-dependent changes in scaling is limited and should be interpreted with caution. By contrast, rearing density showed no consistent effect on scaling relationships. Although slopes varied slightly across density treatments (Figure 3), and interactions between

body size and environmental variables had 95% credible intervals that overlapped zero (Table S1), indicating little evidence of environmentally induced changes in scaling exponents. Environmental effects on intercept terms were also limited and uncertain. Main effects of diet and density overlapped zero, and although there was weak evidence for a sex-by-diet interaction (sex $\times$ diet: $\beta = 0.29$, 95% CrI: -0.01 – 0.58), uncertainty in this estimate was high. As above, intercepts are not directly biologically interpretable because the body size predictor is uncentred.

Morph-specific patterns in males

Analyses separating male morphs (Table S2) showed clear differences in overall forceps size between morphs, with macrolabial males exhibiting larger forceps than brachylabial males (morph: $\beta = 0.26$, 95% CrI: 0.07 – 0.44). Allometric slopes were positive across morphs (logP: $\beta = 0.36$, 95% CrI: 0.14 – 0.59), but there was little evidence for consistent differences in scaling exponents between morphs (logP $\times$ morph: $\beta = 0.08$, 95% CrI: -0.24 – 0.40). Visual patterns suggest that macrolabial males may exhibit greater sensitivity of scaling relationships to diet (Figure 3c), whereas brachylabial males show more parallel scaling relationships across treatments (Figure 3b). However, statistical support for morph-specific differences in slope plasticity was weak, with credible intervals for interactions between body size, morph, and environmental variables overlapping zero (Table S2; Figure 4). Our results indicate that while male morphs differ clearly in overall trait expression, there is limited evidence that they differ consistently in the environmental sensitivity of allometric scaling.

Among-family variation

Variation among families was modest but non-zero (Table S1). The standard deviation of family-level intercepts was low ($\sigma_{\text{Intercept}} = 0.03$, 95% CrI: 0.001 – 0.071), indicating limited variation

among families in overall trait values. There was somewhat greater variation in allometric slopes among families ($\sigma_{\log p} = 0.05$, 95% CrI: 0.004 – 0.126), consistent with the spread of slope estimates shown in Figure 4. The correlation between family-level intercepts and slopes was negative but highly uncertain ($r = -0.21$ 95% CrI: -0.91 – 0.76), providing little evidence for a consistent association between intercepts and scaling among families.

Discussion

Static allometries are often regarded as stable descriptors of morphological architecture (Huxley 1932; Gould 1966; Voje 2016). However, the evolutionary flexibility of scaling relationships remains debated, with some studies suggesting that slopes can evolve yet remain constrained by developmental integration (Harrison 2015). Our findings indicate that allometric slopes are largely stable but can vary modestly with environmental conditions, although this plasticity is uneven. In the European earwig, weapon exaggeration is not fully described by a single static slope, but instead shows patterns consistent with scaling relationships that differ between sexes and vary across developmental environments. Males exhibited steep positive allometry, consistent with expectations for sexually selected weapons (Kelly 2005; Kodric-Brown et al. 2006; Bonduriansky 2007a), and slopes tended to be shallower under nutritional stress. However, uncertainty in several condition-specific estimates indicates that support for slope plasticity is limited and context-dependent. Females, by contrast, showed comparatively stable scaling relationships across environments. Together, these results suggest that the allometric slope is largely stable but may exhibit modest environmental sensitivity, rather than representing a strictly fixed constant (Frankino et al. 2005; Shingleton et al. 2007; Nijhout and German 2012).

303 Accordingly, our study provides evidence that condition dependence may extend beyond
304 variation in mean trait expression to aspects of size-dependent exaggeration itself (Rowe and
305 Houle 1996; Cotton et al. 2004; Tomkins et al. 2004; Pélabon et al. 2014).

306 The evolution of exaggerated traits is often interpreted through the lens of condition dependence
307 and genic capture theory (Rowe and Houle 1996; Cotton et al. 2004; Tomkins et al. 2004), which
308 predicts heightened environmental sensitivity in sexually selected characters. Although many
309 studies have focused on mean trait expression, condition dependence has also been interpreted as
310 variation in allometric slopes (e.g. Bonduriansky 2007b; Vea et al. 2023). However, empirical
311 tests of how environmental conditions shape scaling exponents remain relatively rare. If growth
312 processes are condition-sensitive, the scaling exponent itself may vary across environmental
313 contexts (Emlen 1997; Frankino et al. 2005; Shingleton et al. 2007; Nijhout and German 2012).
314 Our findings provide limited evidence for such slope variation in a weapon trait. In males, the
315 relationship between trait size and body size tended to be less steep under a poor diet, consistent
316 with reduced exaggeration among larger individuals. However, uncertainty in several condition-
317 specific estimates indicates that support for diet-dependent changes in scaling is uneven. In
318 contrast, we found little consistent evidence that rearing density alters scaling relationships.
319 Because the scaling exponent captures how disproportionately trait size covaries with body size,
320 even modest or context-dependent variation in the slope could influence the strength of size-
321 dependent exaggeration and therefore the opportunity for sexual selection to favour larger males.
322 Accordingly, our results are most consistent with the idea that slope variation may be associated
323 with environmentally mediated differences in growth processes, although the magnitude and
324 generality of this effect remain uncertain.

Analyses separating male morphs refine the interpretation of morphological differentiation in earwig weapons. Macrolabial males exhibited the steepest allometric slopes under favourable conditions, but these slopes were reduced—and more variable—under a poor diet, suggesting that condition dependence may be more pronounced in this exaggerated morph. However, statistical support for morph-specific differences in slope plasticity was limited. In contrast, brachylabial males showed more moderate slopes with comparatively little environmental sensitivity. Despite clear differences in trait size between morphs, we found little evidence that morphs differ consistently in intercepts independent of body size. Distinguishing between intercept and slope remains important because these parameters capture different statistical aspects of the relationship between trait size and body size: the slope describes how trait size covaries with body size, whereas the intercept reflects the expected trait size at a given body size (Bonduriansky and Day 2003; Pélabon et al. 2014). Our results therefore suggest that morph differences in exaggeration arise primarily from differences in scaling relationships rather than fixed shifts in expected trait size. This pattern is broadly consistent with models in which exaggerated phenotypes are more sensitive to condition, even when overall scaling relationships are similar (Klingenberg 2008; Hansen and Houle 2008; Buzatto et al. 2015).

Body-size overlap between morphs also provides insight into the basis of morph determination. Although macrolabial males generally possess larger forceps, considerable overlap in body size occurs between morphs, with some small males expressing the exaggerated phenotype and many large males remaining brachylabial (see [Figure 1](#)). If morph expression depended solely on a strict body-size threshold, a sharper transition would be expected. Instead, these observations suggest that body size alone is insufficient to predict morph identity. Previous work has shown that both

environmental and genetic factors influence morph expression in *F. auricularia* (Tomkins 1999). Our findings are consistent with this view: morphs differ strongly in trait expression yet show broadly similar scaling relationships, indicating that morph differentiation does not require fundamentally different scaling patterns (see also Kelly 2024; Kelly 2026). Although we cannot estimate the heritability of morph identity directly, the overlap in body size between morphs and the presence of modest among-family variation in scaling parameters suggest that morph determination is unlikely to be governed by a simple size-based switch.

This hierarchical structure—sex-specific differences in slopes combined with morph-specific differences in scaling sensitivity—suggests that exaggeration may evolve through modifications of scaling relationships across biological levels rather than through uniform shifts in trait expression. Sex-limited expression may allow selection to modify male size–trait relationships without directly affecting females, whereas morph differentiation appears to influence how scaling relationships vary across conditions. Selection may therefore act on different components of scaling across biological levels, particularly on slopes at the sex and morph level.

Among-family variation in scaling relationships was modest but non-zero, indicating that scaling parameters vary slightly across families with shared developmental or genetic backgrounds. However, we found little evidence for a consistent relationship between family-level intercepts and slopes, as estimates of their covariance were highly uncertain. While family effects combine genetic and shared environmental influences, these results suggest that most variation in scaling relationships in this system is structured by sex, morph, and environment rather than strong divergence among families. Nonetheless, the presence of among-family variation indicates that

scaling relationships are not fixed and may still provide a substrate for evolutionary change (Lande 1980; Pélabon et al. 2014).

In summary, these results support a view of static allometry as an evolvable and potentially environmentally responsive trait complex (Pélabon et al. 2014), although the strength of this responsiveness appears limited. Scaling exponents appear to be weakly condition-sensitive descriptors of trait–body size covariation, particularly in males, but the magnitude and consistency of this effect remain uncertain. In contrast, there is little evidence for consistent effects on intercepts, suggesting that variation in exaggeration is more closely associated with differences in scaling than with shifts in expected trait size, albeit with substantial uncertainty in some estimates. Importantly, many condition-specific slope estimates were associated with wide credible intervals that overlapped zero, indicating that support for slope plasticity is context-dependent and should be interpreted cautiously. Accordingly, our results are most consistent with the view that scaling relationships are largely stable but may exhibit modest, environmentally contingent variation. The patterns observed here suggest that the evolution of exaggerated weapons may involve modification of scaling relationships across biological levels, rather than relying solely on changes in average trait size. Our analyses quantify statistical patterns of covariation and do not directly identify the developmental mechanisms underlying these relationships. Future work integrating quantitative genetic estimates of slope variation with experimental manipulations of selection will be essential for determining how readily scaling relationships evolve.

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490 **Figures**

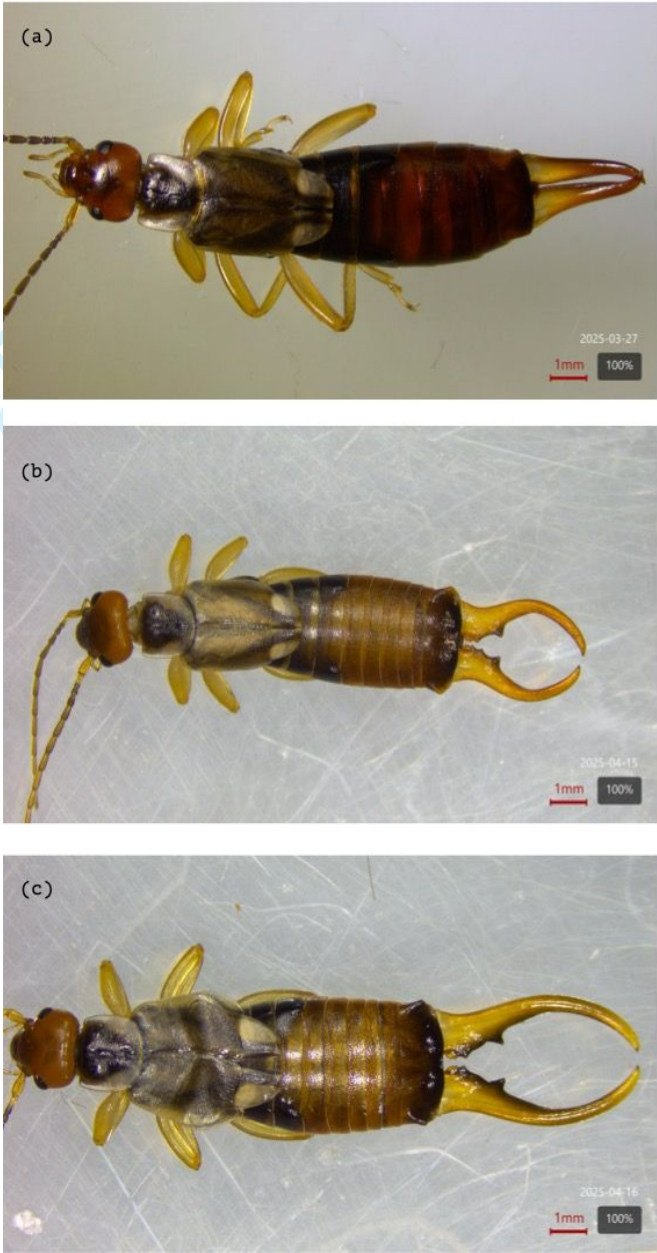


Figure 1: Visual comparison of forceps size and shape across sexes and male morphs: (a) female, (b) brachylabic male, and (c) macrolabic male. Macrolabic males display exaggerated forceps consistent with positive allometry, while brachylabic males and females exhibit relatively reduced forceps. Scale bars = 1 mm.

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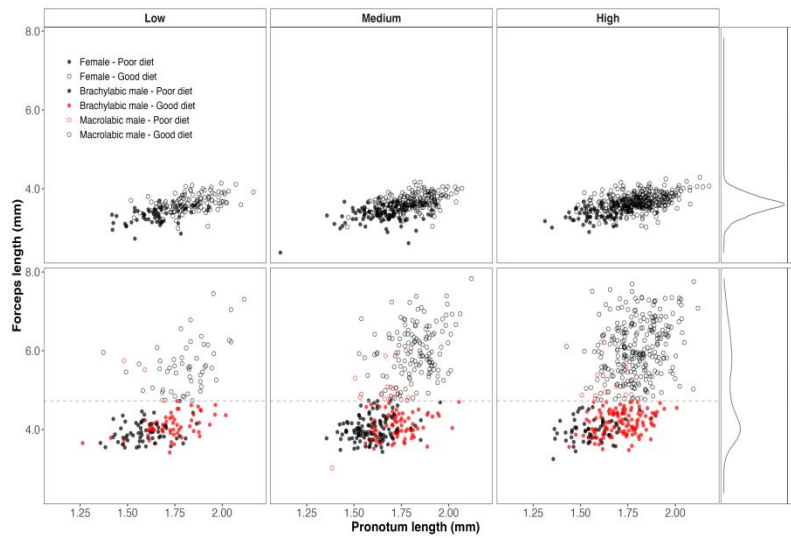


Figure 2: Static allometries of forceps length as a function of pronotum length across density treatments. Females (top row) and males (bottom row) are shown separately. In males, brachylabic (solid points) and macrolabic (open points) morphs are distinguished, with diet treatments indicated by shading. Males form two clusters corresponding to morphotype, with overlap in body size between morphs. The dashed horizontal line in the male panels indicates the forceps length threshold used for morph classification.

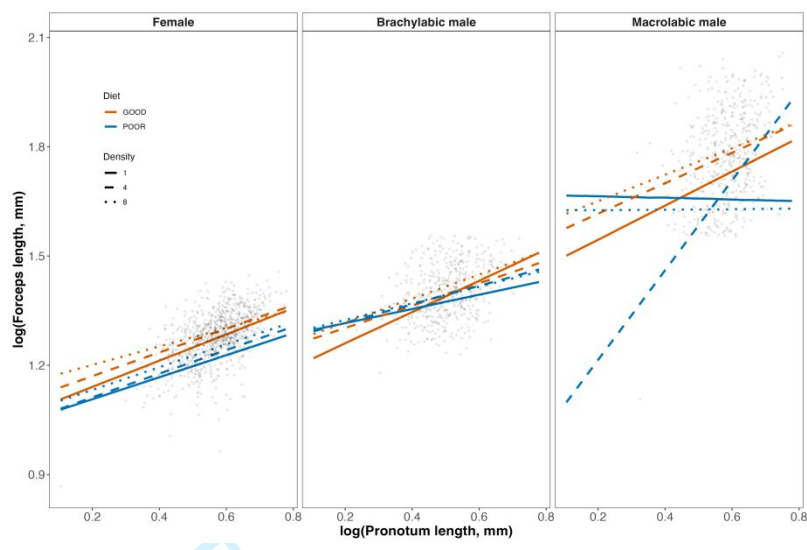


Figure 3: Predicted relationships between body size (log pronotum length) and forceps length across sexes and morphs for each environmental condition combination. Lines show posterior median predictions from the Bayesian model, and points represent raw data.

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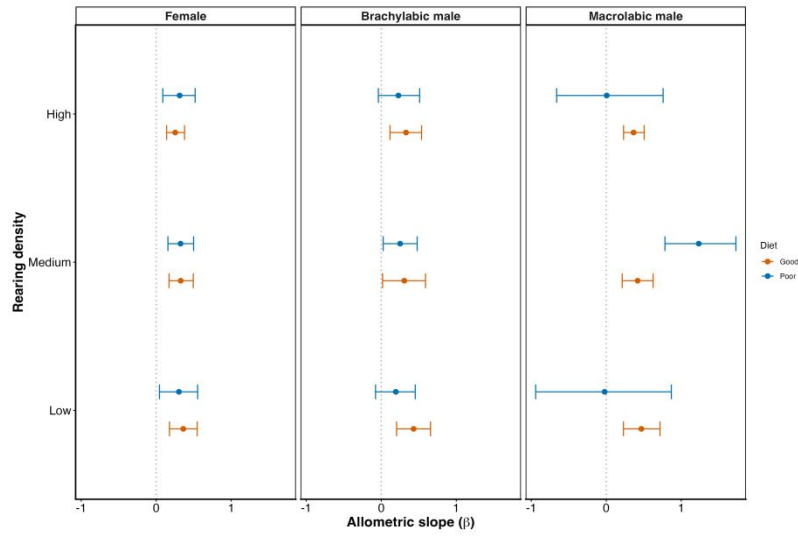
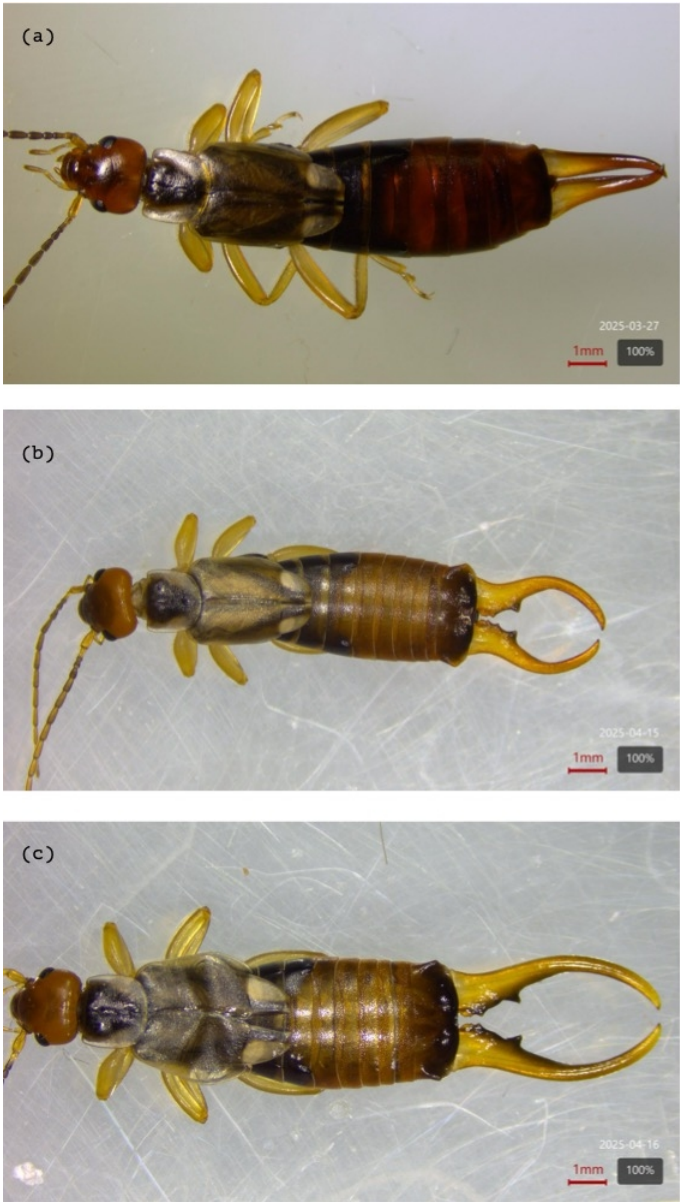


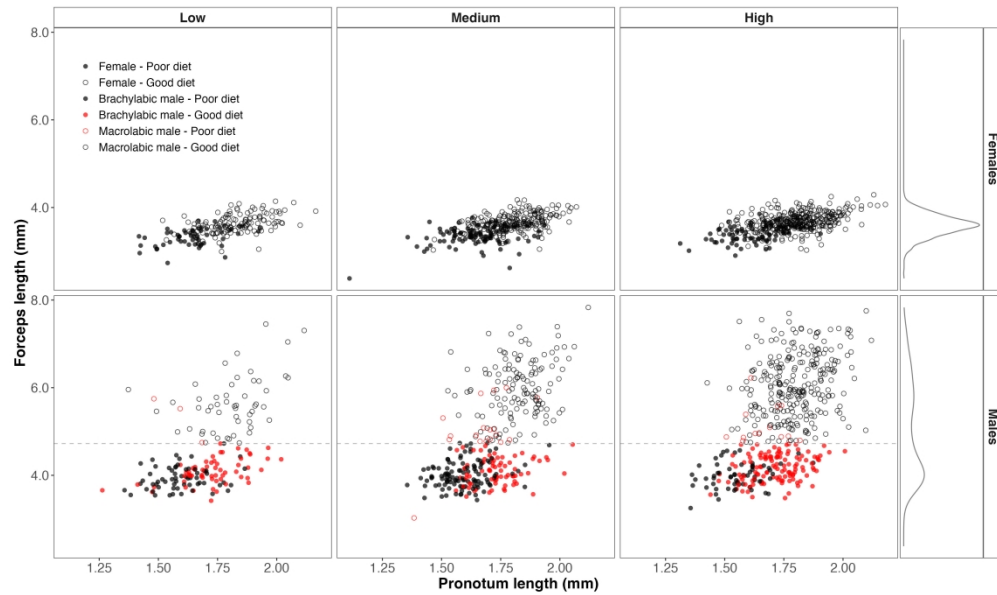
Figure 4: Posterior estimates of allometric slopes (β) across sexes and male morphs under different environmental conditions. Points show posterior medians and horizontal bars indicate 95% credible intervals. Estimates are shown for each combination of diet and density treatments.

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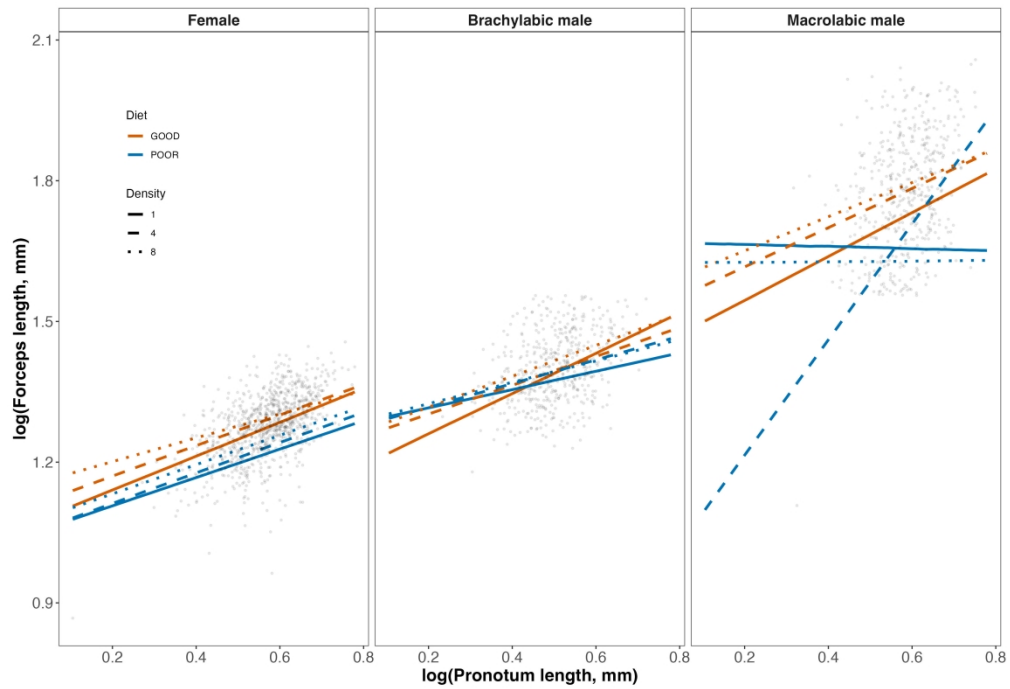
Visual comparison of forceps size and shape across sexes and male morphs: (a) female, (b) brachylabic male, and (c) macrolabic male. Macrolabic males display exaggerated forceps consistent with positive allometry, while brachylabic males and females exhibit relatively reduced forceps. Scale bars = 1 mm.

199x352mm (72 x 72 DPI)



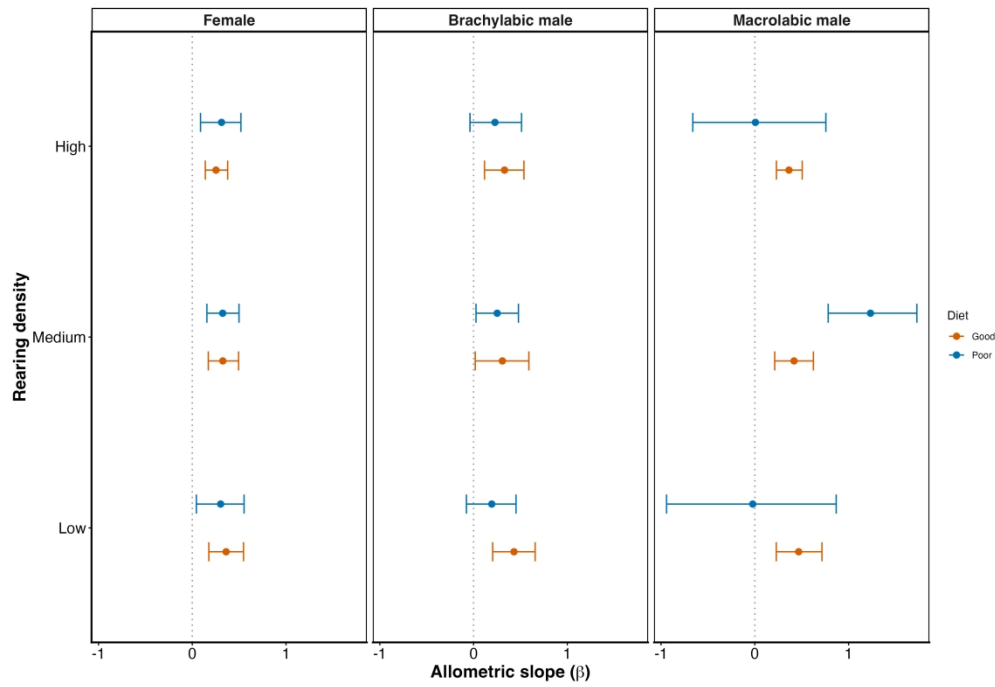
Static allometries of forceps length as a function of pronotum length across density treatments. Females (top row) and males (bottom row) are shown separately. In males, brachylabic (solid points) and macrolabic (open points) morphs are distinguished, with diet treatments indicated by shading. Males form two clusters corresponding to morphotype, with overlap in body size between morphs. The dashed horizontal line in the male panels indicates the forceps length threshold used for morph classification.

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Predicted relationships between body size (log pronotum length) and forceps length across sexes and morphs for each environmental condition combination. Lines show posterior median predictions from the Bayesian model, and points represent raw data.

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Posterior estimates of allometric slopes (β) across sexes and male morphs under different environmental conditions. Points show posterior medians and horizontal bars indicate 95% credible intervals. Estimates are shown for each combination of diet and density treatments.

1357x934mm (72 x 72 DPI)

Abstract

Static allometries are often ~~regarded~~treated as stable descriptors of morphological architecture, yet scaling relationships themselves ~~can~~may evolve and respond plastically to environmental ~~variation~~conditions. We quantified sex- and morph-specific weapon allometries (focusing on the forceps, which are hardened rear appendages used for defence, hunting, and mating) in the European earwig (*Forficula auricularia*) under experimental manipulation of developmental diet and density, ~~while estimating among-family variation in intercepts and using hierarchical models that allow~~ scaling exponents ~~to vary across biological levels~~. Forceps length scaled positively with body size in ~~both sexes~~all groups, but males exhibited substantially steeper allometric slopes ~~under favourable conditions~~. ~~Nutritional~~than females. There was some evidence that nutritional stress was associated with reduced male scaling exponents ~~towards female-like values, whereas increased developmental density amplified male slopes, indicating environmentally sensitive reaction norms of scaling, although support for these effects was variable and context-dependent, while density had little consistent influence on slopes~~. Female scaling relationships remained comparatively stable across environments. ~~In~~Among males, macrolabial ~~and individuals tended to show greater environmental sensitivity in scaling relationships than brachylabial morphs differed primarily in~~ males, although statistical support for these differences was limited. There was little evidence that environmental or morph-specific differences were driven by intercept ~~rather than slope, suggesting divergence in baseline allocation within a shared scaling framework. Among families, intercepts and shifts independent of body size. Among-family variation in slopes~~ ~~covaried positively, indicating that evolutionary change in mean exaggeration would likely alter~~

~~size-dependent exaggeration as a correlated response. Together~~ and intercepts was modest, and
their covariance was weak and uncertain. Overall, these results ~~shows~~suggest that weapon
allometry is structured ~~across sex, morph, environment, and family levels, and that primarily~~
~~through variation in~~ scaling exponents ~~are evolvable and provide qualified support for the idea~~
~~that slopes may represent condition-sensitive~~ components of exaggerated trait architecture.

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Introduction

Static allometry describes how trait size ~~seals~~covaries with body size among individuals at the same developmental stage and provides a central framework for understanding the evolution of exaggerated morphology. Sexually selected weapons often exhibit positive allometry, such that larger individuals ~~invest~~tend to express disproportionately ~~more in trait size~~larger traits (Kodric-Brown et al. 2006; Bonduriansky ~~2007~~2007a; Voje 2016). However, hyperallometry is not universal. When traits serve dual sexual and viability functions, or when functional and mechanical constraints limit disproportionate investment, isometry or hypoallometry may evolve instead (Bonduriansky and Day 2003; Eberhard et al. 2018; Kelly 2026). Consequently, ~~the~~ scaling ~~relationship itself reflects~~relationships reflect how developmental ~~allocation~~processes, functional performance, and selection ~~interact to shape~~are associated with patterns of trait ~~architecture~~variation.

Static allometry is typically described by a linear relationship between log-transformed trait size and body size. In this framework, the slope (scaling exponent) describes the strength of covariation between trait size and body size (i.e., size-dependent exaggeration), whereas the intercept (elevation) represents expected trait size at a reference value of body size (e.g., the mean when body size is centred). Static allometry can therefore diverge ~~due to~~through changes in ~~the scaling exponent or the intercept either parameter~~. Differences in scaling exponent reflect variation in ~~how strongly at~~the strength of covariation between trait ~~responds to~~size and body size, whereas intercept divergence reflects differences in ~~baseline~~expected trait ~~expression~~size at a ~~given~~that reference body size. These components need not evolve together (Lande 1980; Cheverud 1982; Pélabon et al. 2014). Sexual dimorphism, for example, may arise because males

and females differ in scaling sensitivity, in ~~baseline allocation~~expected trait size, or in both. Similarly, alternative male morphs may diverge through elevation shifts while retaining similar scaling exponents, or through ~~fundamentally different~~differences in scaling ~~architectures~~relationships (Tomkins et al. 2005). Distinguishing between slope-based and intercept-based divergence is therefore essential for understanding how exaggerated traits evolve (Bonduriansky ~~2007~~2007a; Shingleton and Frankino 2013; Pélabon et al. 2014; Voje 2016).

Species expressing alternative reproductive phenotypes provide particularly valuable models for examining the structure of allometry. In these species, males adopt discrete morphs that differ in weapon size, behaviour, and mating tactics (Gross 1996; Shuster and Wade 2003; Oliveira et al. 2008). Because morphs share much of their genome and developmental environment, differences in scaling ~~architecture~~parameters can reveal how ~~developmental pathways partition~~ exaggeration varies within a single species. However, empirical analyses of morph-specific static allometry remain scarce, and even fewer studies have investigated how environmental factors influence both slopes and intercepts in polymorphic systems (Kochensparger et al. 2024; Kelly 2026).

~~The European earwig (*Forficula auricularia*) serves as a classic example of male weapon polymorphism. Males express discrete forceps morphs that vary considerably in size and curvature, commonly referred to as macrolabial and brachylabial forms (Simmons and Tomkins 1996; Tomkins et al. 2005). Forceps function both as sexual weapons during male-male interactions and as defensive structures, making them traits with dual roles (Kamimura 2014; Meunier 2024). Previous work has documented morph-specific differences in forceps exaggeration and scaling, suggesting that divergence occurs not only in absolute size but potentially in scaling architecture. Nonetheless, most studies have primarily compared static~~

~~slopes, with less focus on intercept divergence, environmental sensitivity of scaling parameters, and variation among families.~~

Environmental conditions can influence ~~developmental allocation~~trait and body size variation and consequently reshape static allometries (Emlen 1997; Frankino et al. 2005; Shingleton et al.

2007). Nutritional stress may ~~decrease baseline investment in exaggerated traits, lowering~~reduce expected trait size at a given body size, leading to lower intercepts, whereas social factors such as

rearing density could alter ~~competitive allocation~~patterns. Effects of the environmenttrait

expression relative to body size. Environmental effects on scaling exponents would

~~suggest~~indicate plasticity in the ~~trait-to-body size~~relationship between trait size and body size,

whereas consistent slopes across environments would imply canalisation of ~~the~~scaling

~~architecture~~relationships (Frankino et al. 2005; Shingleton et al. 2007; Nijhout and German

2012). Determining whether environmental variation modifies intercepts, slopes, or both is

therefore crucial ~~to~~for evaluating the ~~developmental~~flexibility of trait exaggeration.

Complementary structural modelling of the same study system demonstrates that environmental

effects on forceps size are transmitted both indirectly via body size and directly through trait-

specific ~~allocation~~processes, particularly in males (Ozkan and Kelly submitted).

~~Besides~~Beyond sex, morph, and environment, variation among maternal families of related

individuals offers insights into the ~~integration~~structure of scaling ~~architecture~~relationships.

Intercept and slope may covary across lineages if shared genetic or developmental ~~processes~~

~~jointly regulate baseline allocation~~factors influence both expected trait size and scaling

~~sensitivity~~the strength of covariation with body size (Lande 1980; Hansen and Houle 2008;

Klingenberg 2008; Pélabon et al. 2014). A positive intercept–slope covariance indicates that

families producing relatively larger weapons at average body size also ~~have~~exhibit steeper scaling relationships, reflecting ~~integrated exaggeration architecture~~coordinated variation in trait expression (Bonduriansky and Day 2003; Hansen and Houle 2008; Pélabon et al. 2014).

Conversely, independence between parameters suggests modularity or ~~developmental~~relative decoupling of ~~baseline exaggeration and scaling sensitivity~~these components.

The European earwig (*Forficula auricularia*) provides a powerful system for integrating these perspectives on static allometry. Males express discrete forceps morphs that vary considerably in size and curvature, commonly referred to as macrolablic and brachylablic forms (Simmons and Tomkins 1996; Tomkins et al. 2005) (Figure 1). Forceps function both as sexual weapons during male–male interactions and as defensive structures, making them traits with dual roles (Kamimura 2014; Meunier 2024). Previous work has documented morph-specific differences in forceps exaggeration and scaling, suggesting that divergence occurs not only in absolute size but also potentially in scaling relationships. Nonetheless, most studies have primarily compared static slopes, with less attention to intercept divergence, environmental sensitivity of scaling parameters, and variation among families.

Here, we quantify sex-specific and morph-specific static allometries of forceps length in *F. auricularia* under experimentally manipulated diet and density. Rather than treating the scaling exponent as ~~a fixed trait descriptor~~, we test whether it ~~functions as a condition-sensitive developmental parameter~~structured varies across biological levels and environmental conditions. Specifically, we investigate whether sexual dimorphism in weapon exaggeration results from divergence in the scaling exponent, intercept, or both; whether macrolablic and brachylablic males differ mainly in ~~baseline allocation~~expected trait size or in the ~~sensitivity~~strength of covariation

between forceps ~~growth to size and~~ body size; and whether larval environmental conditions influence intercepts, slopes, or their covariance. Finally, by estimating ~~variance~~ variation among families in intercepts and scaling exponents, we ~~determine~~ assess whether ~~baseline exaggeration and size sensitivity are developmentally integrated~~ these components of scaling relationships covary across shared backgrounds. Through integrating sex, morph, environment, and family components within a unified hierarchical framework, we ~~assess~~ evaluate whether static allometry is best characterised as a stable ~~architectural constant~~ descriptor or as an emergent property of ~~layered growth regulation~~ multiple sources of variation (see also Simmons and Tomkins 1996; Tomkins et al. 2005; Kelly 2026).

Methods

Study species and experimental design

~~Females~~ Female (n = 115) *Forficula auricularia* were hand-collected at night from vegetation in Parc Outremont (Montréal, QC, Canada) during September and October 2024. Individuals were transported to the laboratory at the Université du Québec à Montréal and placed individually in 90-mm Petri dishes filled with moist sand. Each dish was provisioned with a cotton-plugged water tube, and TetraMin Tropical Flakes (United Pet Group, Inc., Germany), and each female was assigned a unique ID code. Earwigs were maintained in the dark at 10 °C in an incubator (Percival Scientific, Perry, IA, USA) to mimic overwintering conditions. Dishes were checked daily for egg laying, and food and water were replenished as needed. Food and water were removed at the time of egg laying (approximately early December), after which dishes were checked weekly. Starting January 2025, dishes were checked daily for egg hatching. The date of

egg hatching was recorded; hatching occurred between 13 January and 29 January 2025, and all 115 females produced hatchlings.

For two weeks following hatching, families were provisioned with a water tube and food (TetraMin Tropical Flakes): crude protein: 46%, crude fat: 11%, crude fibre: 3%. This interval allowed nymphs to complete a moult and reach a size sufficient to withstand manual transfer to experimental dishes. To manipulate the developmental environment, we varied two ecological factors: nutritional condition (diet quality) and social density. Diet treatment consisted of two levels: a “good” diet (100% TetraMin Tropical Flakes), providing high nutritional quality, and a “poor” diet (75% TetraMin Tropical Flakes, 25% alpha-non-digestible α -cellulose), providing reduced nutritional content. Density treatment consisted of three levels: low density (one individual per Petri dish), medium density (4 individuals), and high density (8 individuals). These treatments allowed us to examine both nutritional and social environmental effects on growth and morphology. We used a split-brood experimental design in which offspring were randomly assigned to diet and density treatments within maternal families to minimise confounding between maternal background and environmental treatment. Experimental dishes were maintained at 20°C to mimic post-winter developmental conditions and contained a water tube, treatment-specific food, and a refuge (four small pieces of paper straw glued together). A total of 1,265 Petri dishes were prepared. Females (mothers) were euthanized by freezing at –20 °C after offspring had been assigned to treatments. Dishes were checked several times per week for mortality and sexual maturation, and food and water were replenished as needed.

Morphological measurements

Once offspring eclosed to adulthood, they were removed from their Petri dish, the date was recorded, and they were assigned a unique ID code. Pronotum widthlength (a proxy for overall body size) and forceps length ~~(following Lieshout and Elgar 2009)~~ were photographed under a Leica S6D stereo microscope (Leica Microsystems Inc., Concord, ON, Canada) at 0.68× magnification using Enersight software, which automatically adds a scale bar to each image. Measurements were taken to the nearest 0.001 mm (following Lieshout and Elgar 2009) using Fiji image analysis software (Schindelin et al., 2012). All measurements were taken blind to treatment where possible. To assess repeatability, a subset of 30 individuals was measured twice. Intraclass correlation coefficients indicated high repeatability (pronotum width ICC = 0.94; forceps length ICC = 0.97). Log-transformed pronotum width was mean-centred before analysis to facilitate interpretation of intercepts and reduce collinearity among predictors.

Data preparation

All analyses were conducted in the R (Statistical Environment [version 4.4.2; R Core Team (2025 version XXX)-]). Data were processed using functions from the tidyverse (e.g., dplyr), including recoding, filtering, and grouping operations. We retained individuals from experimental dishes that met predefined density criteria to ~~maintain experimental~~preserve the integrity of ~~the~~ density treatments (low density-~~4~~: one survivor; medium density-~~4~~: ≥3 survivors; high density-~~8~~: ≥6 survivors). Individuals lacking sex information were excluded. Forceps length and pronotum width were log-transformed ~~before analysis~~ to linearize allometric relationships and reduce heteroscedasticity. Log-transformed pronotum widthlength was ~~mean-not~~ centred prior to ~~facilitate interpretation of analysis.~~ As a result, model intercepts represent expected forceps length

~~when log-transformed pronotum width equals zero and reduce collinearity in interaction models.~~
~~GOOD~~are therefore not directly interpretable as trait values at a biologically typical body size.
~~Good~~ diet and ~~low~~ density ~~4~~ were treated as reference levels throughout. Female sex and
brachylabic males served as reference categories where appropriate. Maternal identity was
included as a ~~random effect~~ (grouping factor) in all models ~~to account, with both random~~
~~intercepts and random slopes~~ for ~~shared developmental background among siblings~~body size.

Male morph classification

Male morphs (brachylabic and macrolabic) were classified using finite mixture modelling
~~implemented~~ in the R package *mixsmsn* (~~[version 1.1.12;~~ Prates et al. (2013). ~~Consistent with the~~
~~documented weapon dimorphism in earwigs, we]~~). We modelled the distribution of raw male
forceps length as a two-component skew-normal mixture- ~~using the function~~ *smsn.mix()*. Forceps
length was analysed without prior size correction so that morph classification reflects emergent
weapon phenotype rather than residual variation after ~~accounting for body size~~ ~~correction~~.
Models were estimated using the expectation–maximization algorithm with automatic
initialization (maximum 1000 iterations). ~~Model support~~Support for bimodality was evaluated
using AIC and BIC. Posterior probabilities of component membership were ~~obtained~~calculated
for each male, and individuals were assigned to morphs based on maximum posterior probability.
All ~~males~~individuals were retained for subsequent analyses, ~~thereby~~avoiding arbitrary ~~trait~~
thresholds and preserving variation near the morph boundary.

Sex-specific allometry

~~To test whether males~~ We analysed sex-specific and morph-specific allometries using two
complementary models to address distinct biological questions (see Supplementary Material for

statistical model equations). The sex-specific model evaluates overall differences between females ~~differed~~ and males in ~~foreeps allometry~~ scaling relationships and ~~in the~~ their environmental sensitivity ~~of scaling relationships~~. Because males comprise two discrete morphs that differ in trait expression, this analysis effectively captures average differences between females and the male population as a whole, while allowing slopes to vary across predictors through interaction terms. The morph-specific model, restricted to males, explicitly tests for differences between brachylabic and macrolabic individuals in scaling parameters and their environmental sensitivity. This two-model approach therefore separates broad sex-level comparisons from within-male heterogeneity, avoiding the need to impose a single structure that simultaneously accommodates sex differences and morph variation within males. Results from the sex-specific model should thus be interpreted as averages across male morphs, whereas the morph-specific model resolves variation within males.

To quantify sex differences in allometric scaling and their environmental sensitivity, we fitted a hierarchical Gaussian model ~~with log~~ using the function `brm()` from the *brms* package (via the *cmdstanr* backend) [version 2.23.0; Bürkner (2017)]. Log-transformed forceps length ~~was~~ specified as the response variable. Fixed effects included ~~mean-centred~~ log pronotum ~~width, length~~ (body size), sex, diet, density, and all corresponding interactions. Interactions between body size and other predictors allow the allometric slope (scaling exponent) to vary across sexes and environmental conditions. Main effects and lower-order interactions ~~necessary to test sex~~ describe differences in intercepts ~~and scaling exponents across environmental treatments~~. Interactions between, however, because body size was not centred, intercepts correspond to trait values at log body size equal to zero and are not directly biologically

meaningful. Three-way and environmental variables allowed scaling exponents to vary across conditions, and three-way interactions involving body size were included to capture potential sex-specific environmental sensitivity modulation of scaling differences between males and females relationships. Maternal identity (n = 110 families) was included as a random intercept and random slope for body size, allowing families to vary in baseline forceps size and scaling exponent both intercepts and slopes.

Morph-specific allometry

To examine scaling architecture within males, we fitted a second hierarchical model restricted to males using `brm()`. Log-transformed forceps length was modeled as a function of body size, male morph (macrolabial versus brachylabial), diet, density, and all corresponding interactions. Body size $\times$ morph interactions were tested to assess baseline slope divergence quantify differences in scaling exponents between morphs, and three-way while lower-order terms describe differences in intercepts. Higher-order interactions with environmental factors were evaluated to assess variables allow both slopes and intercepts to vary across conditions in a morph-specific slope plasticity manner. As morph classification was based on absolute forceps expression, this analysis tests whether evaluates differences in scaling architecture among emergent weapon phenotypes differ in scaling architecture without imposing size correction prior to morph assignment. Maternal identity (n = 104 families) was again included as a random intercept and slope for body size.

Priors and model fitting

Models were fitted in a Bayesian framework (*brms*, Bürkner 2017) using multilevel Gaussian regression using *brms*. Weakly informative priors were specified for fixed effects and variance

components to regularize estimates while remaining non-restrictive on the log scale. Models were fit using Hamiltonian Monte Carlo via *cmdstanr*. Convergence and sampling adequacy were verified/assessed using standard diagnostics: (R, effective sample sizes, and trace plots).

Estimation of scaling exponents and plasticity

Condition-specific allometric slopes were derived from posterior draws ~~as linear combinations of regression coefficients using the function *emtrends()* from the *emmeans* package [version 2.0.1; Lenth et al. (2019)].~~ Environmental plasticity in scaling was quantified as ~~the deviation~~ deviations in slope from the reference condition (~~GOOD~~ good diet, low density~~4~~). We report posterior medians and 95% credible intervals for all slopes, intercepts, and ~~slope differences~~ their contrasts. To ~~evaluate biological relevance~~ aid interpretation, we also calculated posterior probabilities that slope ~~changes~~ differences exceeded zero and that their absolute magnitude exceeded 0.1 on the log–log scale. All inference is based on posterior distributions rather than null-hypothesis significance testing.

Results

~~Sexual dimorphism in scaling relationships~~

Forceps length scaled positively with pronotum length ~~in both sexes~~ across all ~~environmental treatments (Figure 1).~~ Females ~~individuals, confirming overall positive allometry (Table S1).~~ Descriptively, females exhibited a continuous distribution of weapon sizes across body sizes ($\bar{x} \pm$ SE pronotum length = 1.77 ± 0.005 mm; $\bar{x} \pm$ SE forceps length = 3.59 ± 0.008 mm, $n = 935$), whereas males formed two discrete clusters corresponding to brachylabic ($\bar{x} \pm$ SE pronotum length = 1.66 ± 0.006 mm; $\bar{x} \pm$ SE forceps length = 4.064 ± 0.014 mm, $n = 491$) and macrolabic

morphs ($\bar{x} \pm \text{SE}$ pronotum length = 1.80 ± 0.006 mm; $\bar{x} \pm \text{SE}$ forceps length = 5.93 ± 0.033 ; $n = 488$), while maintaining the overall positive scaling relationship between body size and forceps length (Figure 2).

Forceps allometrySex differences in scaling relationships

Allometric slopes differed markedly between the sexes (Table S1). Under good diet and low density conditions (reference levels), females exhibited. Because males comprise two morphs with differing scaling relationships, sex-specific results should be interpreted as averages across male phenotypes. Females showed a positive but moderate scaling positive relationship between body size and forceps length and body size (posterior mean slope (logP: $\beta = 0.3335$, 95% CrI: 0.08–0.55). Males showed 0.07–0.65; Figure 3a), whereas males exhibited substantially steeper scaling, as indicated by the positive interaction between body size and sex (log pronotum $\times$ sexmale (logP $\times$ sex: $\beta = 0.5361$, 95% CrI: 0.22–0.8498; Table S1). Under baseline conditions, the implied male slope was 0.86 (95% CrI: 0.64–1.08). This difference divergence is evident in the separation of predicted scaling relationships between sexes (Figure 2) and in posterior slope estimates (Figure 3). The posterior probability that male slopes exceeded female slopes was >0.9995 , indicating pronounced sexual dimorphism in scaling exponent. Sexes also differed strongly in intercept. At mean body size, males had larger forceps than females (sexmale = 0.28, 95% CrI: 0.26–0.31; Table S1), demonstrating clear size dimorphism independent of slope differences. Thus, sexual dimorphism in weapon exaggeration reflects divergence in both scaling exponent and mean trait size.

Sex-specific environmental sensitivity

Intercept plasticity

Environmental effects on intercept were sex biased. Poor diet reduced forceps size at mean body size in females (dietPOOR = -0.06, 95% CrI: -0.08–0.03; Table S1). This reduction was substantially stronger in males, as indicated by the negative sex × diet interaction (sexmale:dietPOOR = -0.10, 95% CrI: -0.13–0.06; Table S1), yielding a total male intercept decline of approximately -0.15 under poor diet.

Density effects on female intercepts were weak. In contrast, males exhibited positive intercept shifts at higher densities (sexmale:density4 = 0.05; sexmale:density8 = 0.08, both credible intervals excluding zero; Table S1). Thus, males were more environmentally sensitive than females in mean weapon size, showing both nutritional suppression and density-associated increases in exaggeration.

Slope plasticity

Female slopes were relatively stable across environments. The interaction between body size and diet was centred near zero (log pronotum:dietPOOR = -0.01, 95% CrI: -0.24–0.23; Table S1), and density interactions similarly overlapped zero. Female slopes were relatively stable across environments (Figure 4).

In contrast, males exhibited pronounced environment-dependent changes in scaling exponent (Figure 4). Under a poor diet, the male slope declined substantially, as indicated by the negative three-way interaction (log pronotum × sexmale × dietPOOR = -0.53, 95% CrI: -0.88–0.19; Table S1). This corresponds to a reduction from approximately 0.86 under a good diet to approximately

0.33 under a poor diet. Posterior probability that male slope decreased under a poor diet exceeded 0.999, and the probability that the magnitude of change exceeded 0.1 was > 0.993 .

Density had the opposite effect, increasing male slopes at densities 4 and 8 ($\log \text{pronotum} \times \text{sexmale} \times \text{density4} = 0.40$; $\log \text{pronotum} \times \text{sexmale} \times \text{density8} = 0.40$; both credible intervals excluding zero). Posterior probabilities that male slope increases exceeded 0.1 were > 0.952625 . Across treatments, male slope plasticity exceeded female slope plasticity with high probability (> 0.975).

Together, these results demonstrate pronounced male-specific plasticity in scaling exponent, whereas female scaling relationships appear comparatively canalized.

Morphotype differences among males

Given the strong male-specific plasticity, we next examined scaling architecture within males.

Baseline morph differences

Under good diet and low density conditions, brachylabial males exhibited a slope of 0.35 (95% CrI: 0.13–0.57; Table S2). Macrolabial males showed a nearly identical baseline slope ($\log \text{pronotum} : \text{morphmacrolabial} = 0.36$ (95% CrI: -0.31–0.33; Table S2), with posterior probability of a positive difference equal to 0.521. Thus, there was little evidence of baseline slope divergence between morphs. Baseline slope similarity between morphs is evident in predicted scaling relationships (Figure 2) and posterior slope estimates (, where male scaling relationships are consistently steeper than those of females across conditions. There was little evidence for consistent differences in intercept terms associated with sex (sex: $\beta = -0.06$, 95% CrI: -0.28 – 0.15). However, because body size was not centred, intercepts correspond to expected trait

values at log body size equal to zero and are not directly biologically interpretable. Accordingly, differences between the sexes are best understood in terms of divergence in scaling slopes rather than shifts in intercepts.

Including a quadratic term for body size provided no meaningful improvement in model fit (see Supplementary Material for statistical model equation). Although the quadratic model showed a slightly higher expected log predictive density, the difference was small relative to its standard error ($\Delta\text{elpd} = 3.7 \pm 3.8$). In addition, credible intervals for the quadratic term overlapped zero, indicating no clear evidence of curvature in the size–trait relationship. We therefore retained the linear model for all subsequent analyses.

Environmental effects on scaling

Environmental effects on allometry were generally weak. In females, scaling relationships were largely parallel across diet and density treatments (Figure 3a), and interactions between body size and environmental variables had 95% credible intervals that overlapped zero (Table S1), indicating little evidence of environmentally induced changes in scaling exponents. In males, there was some indication that the nutritional environment may influence scaling relationships. The interaction between body size, sex, and diet was negative ($\log P \times \text{sex} \times \text{diet}$: $\beta = -0.68$, 95% CrI: $-1.25 - -0.11$), consistent with a reduction in the steep male allometric slope under poor dietary conditions. This pattern is visible in Figure 3c, where slopes under a poor diet appear shallower than under a good diet, particularly at larger body sizes. However, credible intervals for several condition-specific slopes and derived contrasts overlapped zero (Table S1), indicating that support for diet-dependent changes in scaling is limited and should be interpreted with caution. By contrast, rearing density showed no consistent effect on scaling relationships.

Although slopes varied slightly across density treatments (Figure 3), and interactions between body size and environmental variables had 95% credible intervals that overlapped zero (Table S1), indicating little evidence of environmentally induced changes in scaling exponents. Environmental effects on intercept terms were also limited and uncertain. Main effects of diet and density overlapped zero, and although there was weak evidence for a sex-by-diet interaction (sex $\times$ diet: $\beta = 0.29$, 95% CrI: -0.01 – 0.58), uncertainty in this estimate was high. As above, intercepts are not directly biologically interpretable because the body size predictor is uncentred.

Morph-specific patterns in males

Analyses separating male morphs (Table S2) showed clear differences in overall forceps size between morphs, with macrolabial males exhibiting larger forceps than brachylabial males (morph: $\beta = 0.26$, 95% CrI: 0.07 – 0.44). Allometric slopes were positive across morphs (logP: $\beta = 0.36$, 95% CrI: 0.14 – 0.59), but there was little evidence for consistent differences in scaling exponents between morphs (logP $\times$ morph: $\beta = 0.08$, 95% CrI: -0.24 – 0.40). Visual patterns suggest that macrolabial males may exhibit greater sensitivity of scaling relationships to diet (Figure 3).

Morphs differed strongly in intercept. At mean body size, macrolabial males had substantially larger forceps than brachylabial males (macrolabial = 0.31, 95% CrI: 0.28 – 0.35; Table S2). Morph divergence in weapon size is therefore primarily elevation-based rather than slope-based.

Morph-specific environmental sensitivity

Diet had little effect on intercept in brachylabial males (dietPOOR = -0.01, 95% CrI: -0.04 – 0.01; Table S2), but reduced intercept in macrolabial males (morphmacrolabial:dietPOOR = -0.10, 95%

CrI: -0.15–0.06; Table S2). Nutritional stress, therefore, disproportionately reduced weapon size in the exaggerated morph.

Both morphs showed some tendency toward reduced scaling exponents under a poor diet. In brachylabie males, slope decline was modest ($\log \text{pronotum}:\text{dietPOOR} = -0.11$, 95% CrI: -0.35–0.13; Table S2). In macrolabie males, the additional diet-dependent slope change in macrolabie males was positive but highly uncertain ($\log \text{pronotum}:\text{morphmacrolabie}:\text{dietPOOR} = 0.18$, 95% CrI: -0.31–0.66; Table S2). Posterior contrasts indicated that diet-induced slope plasticity tended to be greater in macrolabie than brachylabie males (posterior probability ≈ 0.60), although credible intervals were broad.

Density-induced slope changes were comparatively weak and uncertain in both morphs, and there was little evidence that density altered scaling exponents differently between morphs. Morph-specific changes in allometric slope relative to baseline conditions are shown in Figure 5.

Overall, in males, morphological differentiation primarily reflects shifts in intercepts rather than fundamentally distinct scaling exponents.

Among-family variation

At the sex level, there was detectable among-maternal-family variance in scaling exponent ($\text{sd}(\log \text{pronotum}) = 0.13$, 95% CrI: 0.01–0.25) and smaller variance in intercept ($\text{sd}(\text{Intercept}) = 0.03$). The intercept–slope correlation was weak and uncertain ($r = 0.17$, CrI overlapping zero), suggesting limited evidence for integrated family-level variation across sexes (Table S2).

Within males, family-level variation in slope was larger ($\text{sd}(\log \text{pronotum}) = 0.19$, 95% CrI: 0.10–0.28). Intercepts and slopes were positively correlated among families ($r = 0.59$, 95% CrI:

0.05–0.93), indicating that broods producing sons with larger forceps at mean body size also tended to exhibit steeper scaling relationships.

Because females were field-collected and potentially multiply mated, these variance components reflect combined genetic and shared environmental effects. Nonetheless, the positive covariance demonstrates integrated brood-level variation in exaggeration architecture.

Summary

Weapon allometry exhibits a hierarchical structure. Sexual dimorphism is characterized by steeper and more environmentally sensitive scaling in males relative to females. In males, morphological divergence is primarily intercept-based, with macrolabial males consistently expressing larger weapons and greater sensitivity to nutritional stress at the intercept. Slope plasticity is pronounced at the sex level and detectable, though weaker and more uncertain, within males, while baseline slopes are similar across morphs. Finally, male scaling intercepts and exponents covary positively among maternal families, indicating integrated clutch-level variation in exaggeration architecture.

c), whereas brachylabial males show more parallel scaling relationships across treatments (Figure 3b). However, statistical support for morph-specific differences in slope plasticity was weak, with credible intervals for interactions between body size, morph, and environmental variables overlapping zero (Table S2; Figure 4). Our results indicate that while male morphs differ clearly in overall trait expression, there is limited evidence that they differ consistently in the environmental sensitivity of allometric scaling.

Among-family variation

Variation among families was modest but non-zero (Table S1). The standard deviation of family-level intercepts was low ($\sigma_{\text{Intercept}} = 0.03$, 95% CrI: 0.001 – 0.071), indicating limited variation among families in overall trait values. There was somewhat greater variation in allometric slopes among families ($\sigma_{\text{slope}} = 0.05$, 95% CrI: 0.004 – 0.126), consistent with the spread of slope estimates shown in Figure 4. The correlation between family-level intercepts and slopes was negative but highly uncertain ($r = -0.21$ 95% CrI: -0.91 – 0.76), providing little evidence for a consistent association between intercepts and scaling among families.

Discussion

Static allometries are often regarded as stable descriptors of morphological architecture (Huxley 1932; Gould 1966; Voje 2016). However, the evolutionary flexibility of scaling relationships remains debated, with some studies suggesting that slopes can evolve ~~while still being~~ yet remain constrained by developmental integration (Harrison 2015). Our findings ~~challenge the view of a fixed scaling exponent by demonstrating~~ indicate that ~~the allometric slope itself~~ slopes are largely stable but can act as a condition-dependent developmental parameter structured across biological levels. ~~vary modestly with environmental conditions, although this plasticity is uneven.~~ In the European earwig, weapon exaggeration is not ~~defined~~ fully described by a single static slope, but ~~by reaction norms of~~ instead shows patterns consistent with scaling relationships that differ between sexes, ~~and~~ vary across developmental environments, ~~and covary among families.~~ Males exhibited steep positive allometry ~~under favourable conditions~~, consistent with expectations for sexually selected weapons (Kelly 2005; Kodric-Brown et al. 2006; Bonduriansky 2007), ~~yet their~~

~~scaling exponents declined toward female-like values~~2007a), and slopes tended to be shallower
 under nutritional stress ~~and increased under higher developmental densities.~~ However,
~~uncertainty in several condition-specific estimates indicates that support for slope plasticity is~~
~~limited and context-dependent.~~ Females, by contrast, showed comparatively stable scaling
 relationships across environments. ~~These Together, these~~ results ~~indicatesuggest~~ that the
 allometric slope is ~~not~~largely stable but may exhibit modest environmental sensitivity, rather than
~~representing a strictly fixed architectural constant but an environmentally responsive parameter of~~
~~growth regulation~~ (Frankino et al. 2005; Shingleton et al. 2007; Nijhout and German 2012). ~~By~~
~~demonstrating environmentally sensitive scaling exponents~~ Accordingly, our study ~~extends~~
~~theories of~~provides evidence that condition dependence may extend beyond variation in mean
 trait expression to ~~the structure aspects~~ of size-dependent exaggeration itself (Rowe and Houle
 1996; Cotton et al. 2004; Tomkins et al. 2004; Pélabon et al. 2014).

The evolution of exaggerated traits is often interpreted through the lens of condition dependence
 and genic capture theory (Rowe and Houle 1996; Cotton et al. 2004; Tomkins et al. 2004), which
 predicts heightened environmental sensitivity ~~of~~in sexually selected characters. Although ~~most~~
~~empirical work has~~ many studies have focused on mean trait expression, ~~theory does not restrict~~
 condition dependence ~~to shifts in intercepts~~ has also been interpreted as variation in allometric
slopes (e.g. Bonduriansky 2007b; Vea et al. 2023). However, empirical tests of how
environmental conditions shape scaling exponents remain relatively rare. If growth ~~regulation~~
~~is~~processes are condition--sensitive, the scaling exponent itself ~~should respond to~~ may vary across
 environmental ~~inputs~~contexts (Emlen 1997; Frankino et al. 2005; Shingleton et al. 2007; Nijhout
 and German 2012). Our findings provide ~~direct~~limited evidence for such slope ~~plasticity~~variation

in a weapon trait. ~~The size–~~In males, the relationship between trait ~~relationship becomessize and~~
body size tended to be less steep under a poor diet, ~~indicating consistent with~~ reduced
 exaggeration among larger ~~males, whereas higher density steepens the relationship, amplifying~~
~~exaggeration among large~~ individuals. However, uncertainty in several condition-specific
estimates indicates that support for diet-dependent changes in scaling is uneven. In contrast, we
found little consistent evidence that rearing density alters scaling relationships. Because the
 scaling exponent ~~determinescaptures~~ how disproportionately trait size ~~increasescovaries~~ with
 body size, ~~plasticityeven modest or context-dependent variation~~ in the slope ~~directly alterscould~~
influence the strength of size-dependent exaggeration and therefore the opportunity for sexual
 selection to favour larger males. ~~Structural equation modelling of the same experimental~~
~~population further indicates that nutritional effects on male weapon size are mediated primarily~~
~~through body size, whereas density effects act more directly on trait allocation (Ozkan and Kelly~~
~~submitted).~~ Together, these results suggest that slope plasticity reflects amplification of shared
 developmental pathways combined with socially responsive allocationAccordingly, our results
are most consistent with the idea that slope variation may be associated with environmentally
mediated differences in growth processes, although the magnitude and generality of this effect
remain uncertain.

~~The absence of clear slope divergence between brachylabial and macrolabial males~~
~~refines~~Analyses separating male morphs refine the interpretation of morphological differentiation
 in earwig weapons. ~~Although macrolabial males express substantially larger forceps at a given~~
~~body size, their scaling exponent is similar to that of brachylabial males, indicating that morph~~
~~divergence arises primarily through elevation rather than slope differences. Tomkins (2005)~~

~~however found that macrolabial males are positively~~ Macrolabial males exhibited the steepest
allometric ~~while brachylabials are isometric~~ slopes under favourable conditions, but these slopes
were reduced—and more variable—under a poor diet, suggesting that condition dependence may
be more pronounced in this exaggerated morph. However, statistical support for morph-specific
differences in slope plasticity was limited. In contrast, brachylabial males showed more moderate
slopes with comparatively little environmental sensitivity. Despite clear differences in trait size
between morphs, we found little evidence that morphs differ consistently in intercepts
independent of body size. Distinguishing between intercept and slope ~~is~~ remains important
because these parameters ~~reflect~~ capture different statistical aspects of ~~growth regulation: the~~
~~relationship between trait size and body size: the~~ slope describes how trait size ~~responds to~~
~~variation in~~ covaries with body size, whereas ~~the~~ intercept reflects ~~baseline allocation~~ the expected
~~trait size~~ at a given body size (Bonduriansky and Day 2003; Pélabon et al. 2014). ~~Elevation shifts~~
~~without major changes in the scaling exponent~~ Our results therefore suggest ~~modification of~~
~~allocation magnitude within a largely shared scaling framework that morph differences in~~
~~exaggeration arise primarily from differences in scaling relationships rather than fixed shifts in~~
~~expected trait size. This pattern is broadly consistent with models in which exaggerated~~
~~phenotypes are more sensitive to condition, even when overall scaling relationships are similar~~
(Klingenberg 2008; Hansen and Houle 2008). ~~This pattern is consistent with environmental~~
~~threshold models of conditional strategy evolution (Hazel et al. 1990; Tomkins and Hazel 2007;~~
Buzatto et al. 2015), ~~in which discrete phenotypes arise once condition exceeds a developmental~~
~~threshold while underlying size-dependent growth relationships remain broadly conserved.~~
Developmental models further show that such thresholds can emerge from nonlinear regulation

of growth pathways (Douglas J. Emlen 2000). Our results, therefore, suggest that weapon dimorphism in the European earwig primarily reflects divergence in baseline allocation rather than the evolution of distinct allometric programmes. Similar patterns occur across taxa, where exaggerated sexually selected traits often retain comparable scaling relationships despite substantial differences in mean exaggeration (Painting et al. 2024).

Body-size overlap between morphs also provides insight into the developmental basis of morph determination. Although macrolabial males generally possess larger forceps, considerable overlap in body size occurs between morphs, with some small males expressing the exaggerated phenotype and many large males remaining brachylabial (see Figure 1). If morph expression depended solely on a strict body-size threshold, a sharper transition would be expected. Instead, these observations suggest that body size alone is insufficient to predict morph identity. Previous work has shown that both environmental and genetic factors influence morph expression in *F. auricularia* (Tomkins 1999). Our findings are consistent with this view: morphs sharediffer strongly in trait expression yet show broadly similar scaling exponents but differ substantially in intercept, implying partially distinct developmental pathways operating within a commonrelationships, indicating that morph differentiation does not require fundamentally different scaling frameworkpatterns (see also Kelly 2024; Kelly 2026). Although we cannot estimate the heritability of morph identity directly, the overlap in body size between morphs and the presence of modest among-family variation in scaling parameters suggest that morph determination is unlikely to be governed by a simple size-based switch.

This hierarchical structure—sex-specific slope plasticity differences in slopes combined with morph-specific intercept divergence—indicates that differences in scaling sensitivity—suggests

that exaggeration may evolve through ~~layered~~ modifications of scaling ~~architecture~~ relationships across biological levels rather than through uniform shifts in trait expression. Sex-limited expression ~~allows~~ may allow selection to modify ~~the~~ male size–trait ~~reaction norm~~ relationships without ~~constraining female growth pathways~~ directly affecting females, whereas morph differentiation ~~primarily alters allocation levels while maintaining a shared scaling framework~~ appears to influence how scaling relationships vary across conditions. Selection may therefore act on different components of scaling ~~architecture at different~~ across biological levels, ~~modifying particularly on~~ slopes at the sex ~~level, intercepts at the~~ and morph level, ~~and covariance patterns among families~~.

The ~~positive covariance between intercept and slope among maternal families within males provides insight into the integration of scaling components and their potential evolvability. Families producing sons with relatively larger forceps at average body size also tend to exhibit steeper scaling exponents~~ Among-family variation in scaling relationships was modest but non-zero, indicating ~~coordinated variation in baseline allocation and size sensitivity. Because intercept and slope covary positively, evolutionary change in mean exaggeration could simultaneously increase the degree of size-dependent exaggeration. Although~~ that scaling parameters vary slightly across families with shared developmental or genetic backgrounds. However, we found little evidence for a consistent relationship between family-level intercepts and slopes, as estimates of their covariance were highly uncertain. While family effects combine genetic and shared environmental influences, these results suggest that most variation in scaling relationships in this system is structured by sex, morph, and environment rather than strong divergence among families. Nonetheless, the presence of ~~structured covariance demonstrates~~ among-family variation

indicates that scaling ~~architecture itself varies among lineages rather than being~~
~~developmentally relationships are not~~ fixed. ~~Such integrated variation and~~ may still provide the
~~raw material a~~ substrate for evolutionary change ~~in the intensity of exaggeration~~ (Lande 1980;
Pélabon et al. 2014).

In summary, these results support a view of static allometry as an evolvable and potentially
environmentally responsive trait complex (Pélabon et al. ~~2014~~-2014), although the strength of
this responsiveness appears limited. Scaling exponents ~~behave as~~ appear to be weakly condition-
sensitive ~~reaction norms that differ between sexes and vary among families~~. Morph
~~differentiation in males mainly reflects~~ descriptors of trait–body size covariation, particularly in
males, but the magnitude and consistency of this effect remain uncertain. In contrast, there is
little evidence for consistent effects on intercepts, suggesting that variation in exaggeration is
more closely associated with differences in scaling than with shifts in ~~baseline allocation,~~
~~whereas the intercept–slope covariance reveals coordinated variation linking baseline~~
~~exaggeration and size sensitivity~~. Together, these patterns indicate expected trait size, albeit with
substantial uncertainty in some estimates. Importantly, many condition-specific slope estimates
were associated with wide credible intervals that overlapped zero, indicating that support for
slope plasticity is context-dependent and should be interpreted cautiously. Accordingly, our
results are most consistent with the view that scaling relationships are largely stable but may
exhibit modest, environmentally contingent variation. The patterns observed here suggest that the
evolution of exaggerated weapons may ~~proceed through hierarchical~~ involve modification of
~~growth regulation~~ scaling relationships across ~~sexes, morphs, and lineages~~ biological levels, rather
than ~~through simply~~ relying solely on changes in average trait size. Our analyses quantify

statistical patterns of covariation and do not directly identify the developmental mechanisms
underlying these relationships. Future work integrating quantitative genetic estimates of slope
variation with experimental manipulations of selection will be essential for determining how
readily scaling architecturesrelationships evolve.

Data availability

~~Data are available through the OSF repository at~~
~~https://osf.io/m7c53/overview?view_only=ce50e42b29284cbe8164e768ecb9c2b1.~~

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683 **Figures**

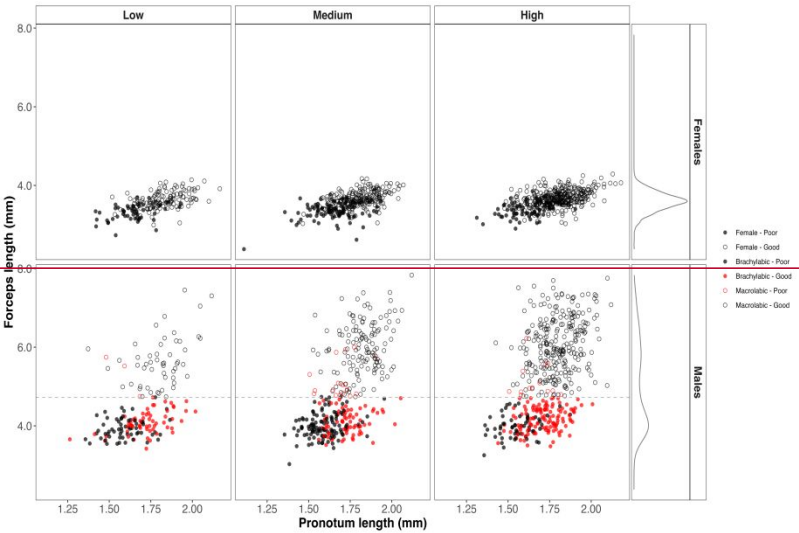


Figure 1: Static allometries of forceps length as a function of pronotum length across density



treatments (low, medium, high).

Figure 1: Visual comparison of forceps size and shape across sexes and male morphs: (a) female, (b) brachylabial male, and (c) macrolabial male. Macrolabial males display exaggerated forceps consistent with positive allometry, while brachylabial males and females exhibit relatively reduced forceps. Scale bars = 1 mm. Females (top row) and males (bottom row) are shown separately. In males, brachylabial (solid points) and macrolabial (open points) morphs are distinguished, with diet treatments indicated by shading. Males form two discrete clusters

corresponding to morphotype, with substantial overlap in body size between morphs. Environmental treatments shift forceps length within morphs but do not eliminate size overlap. The dashed horizontal line in the male panels indicates the approximate threshold commonly used to distinguish morphs based on forceps length.

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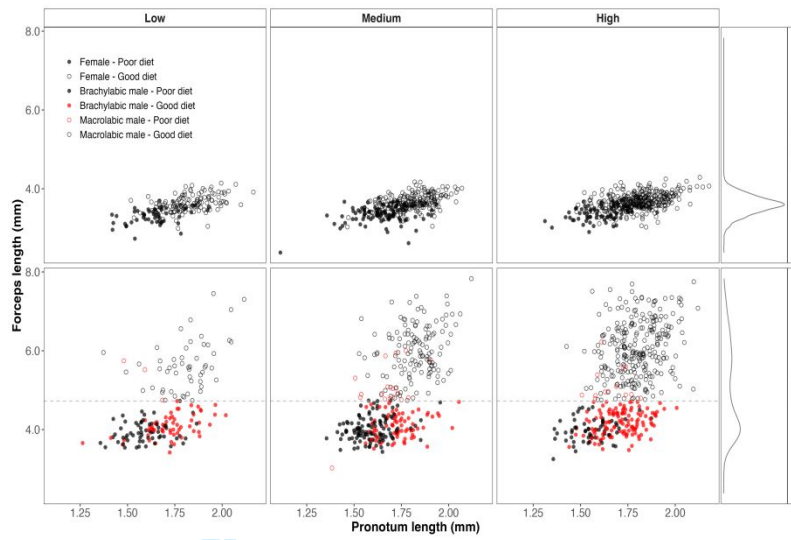
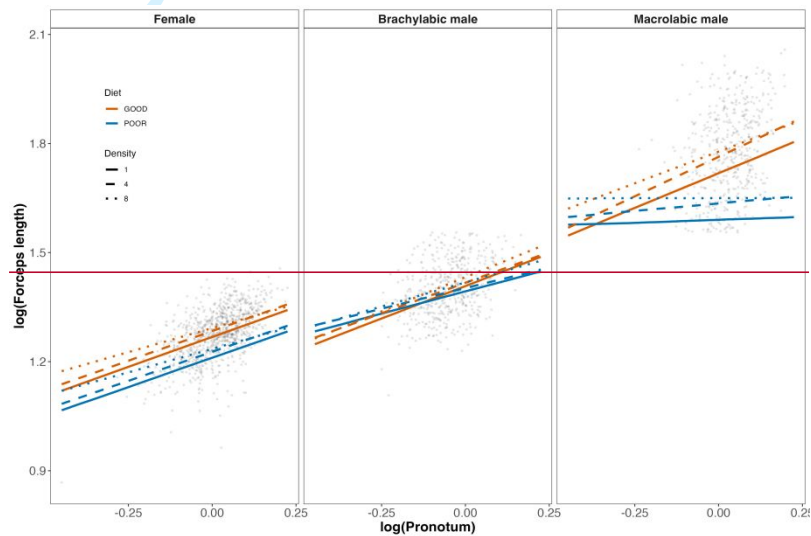


Figure 2: Static allometries of forceps length as a function of pronotum length across density treatments. Females (top row) and males (bottom row) are shown separately. In males, brachylabial (solid points) and macrolabial (open points) morphs are distinguished, with diet



treatments indicated by shading.

Figure 2: Predicted population-level relationships between log-transformed forceps length and log-transformed pronotum width across diet (GOOD vs POOR) and rearing density (1, 4, or 8 individuals per container). Panels show females (left), brachylabial males (center), and macrolabial males (right). Solid, dashed, and dotted lines correspond to density treatments; color indicates diet. Fitted lines represent posterior medians from Bayesian multilevel models excluding random family effects; grey points show raw data. Female slopes are relatively

stable across environments, whereas male slopes exhibit pronounced environmental sensitivity. Morph divergence in males is primarily intercept-based, with macrolabial males exhibiting consistently larger forceps at a given body size and stronger slope reductions under poor diet. Males form two clusters corresponding to morphotype, with overlap in body size between morphs. The dashed horizontal line in the male panels indicates the forceps length threshold used for morph classification.

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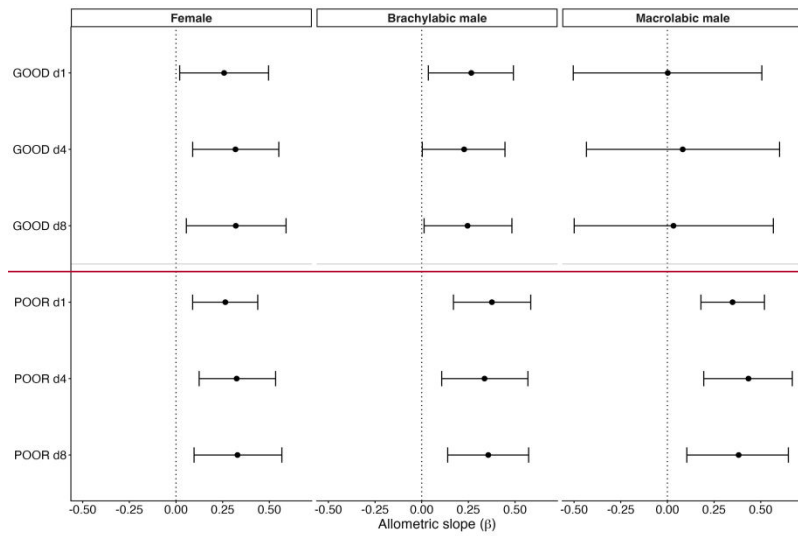


Figure 3: Points represent posterior medians and horizontal bars indicate 95% credible intervals for slopes relating log forceps length to log pronotum width. Treatments are grouped by diet (good vs. poor) and density (low, medium, or high). The vertical dotted line indicates a slope of zero. Female slopes cluster tightly and show little environmental sensitivity. In contrast, male slopes vary substantially across environments, with macrolabic males showing a higher posterior probability of substantial slope reduction under poor diet than brachylabic males.

These estimates are derived from Bayesian multilevel models that incorporate among-family

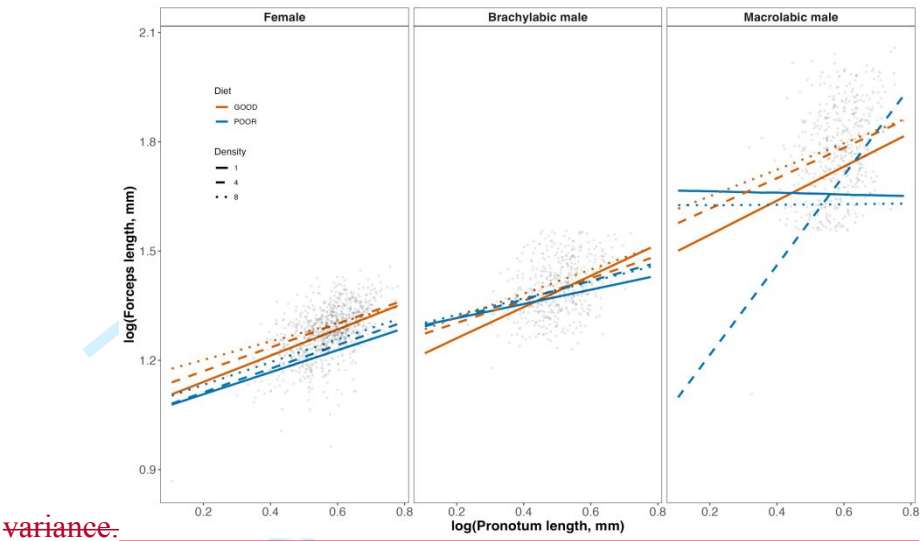


Figure 3: Predicted relationships between body size (log pronotum length) and forceps length across sexes and morphs for each environmental condition combination. Lines show posterior median predictions from the Bayesian model, and points represent raw data.

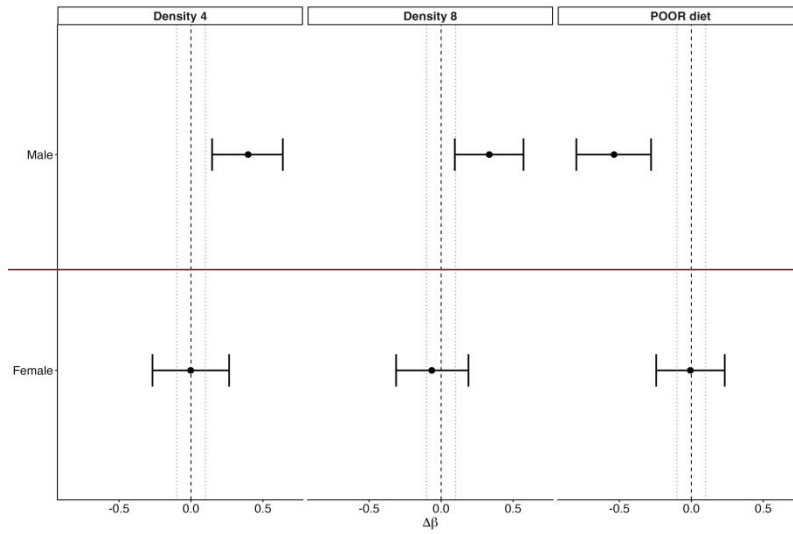


Figure 4: Posterior medians (points) and 95% credible intervals (horizontal bars) show changes in the allometric slope ($\Delta\beta$) of log forceps length on log pronotum width relative to the baseline condition (good diet, low density). The vertical dashed line indicates no change in slope, and the dotted lines denote a biologically meaningful threshold of $|\Delta\beta| = 0.1$. Females show little evidence of slope plasticity across environments. In contrast, males exhibit pronounced environmental sensitivity, with strong reductions in the scaling exponent under poor diet and substantial increases at elevated densities. Estimates are derived from the sex-specific

multilevel model including sex × environment × body size interactions.

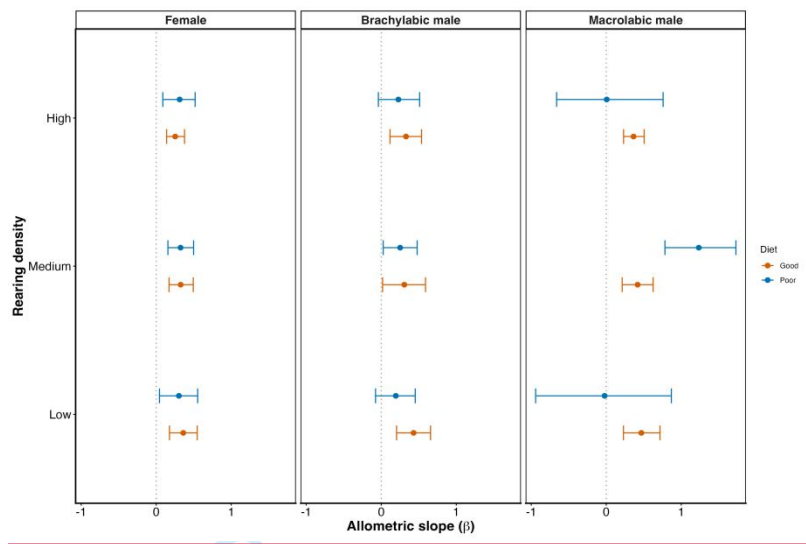


Figure 4: Posterior estimates of allometric slopes (β) across sexes and male morphs under different environmental conditions. Points show posterior medians and horizontal bars indicate 95% credible intervals. Estimates are shown for each combination of diet and density treatments.

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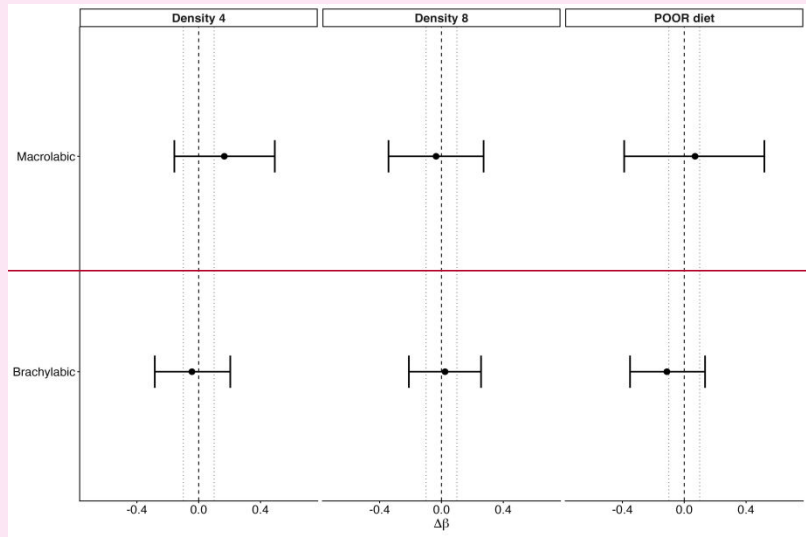


Figure 5: Posterior medians and 95% credible intervals show changes in the allometric slope ($\Delta\beta$) relative to brachylab males under good diet and low density. The vertical dashed line

indicates no change in slope; dotted lines denote $|\Delta\beta| = 0.1$. Both morphs show environmentally induced variation in slope, but the broad overlap of credible intervals provides limited evidence for strong morph-specific divergence in scaling plasticity. Morph differentiation in weapon exaggeration therefore reflects primarily intercept shifts rather than fundamentally distinct scaling exponents. Estimates are derived from the male-only multilevel model including morph $\times$ environment $\times$ body size interactions.

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